

valid looking proofs for false statements.

This makes security auditing of ZKP-based systems extremely important and complex. Unlike regular software bugs, ZKP vulnerabilities can be subtle and mathematically non-obvious.

Quantum computing threat:

zkSNARKs rely on the hardness of elliptic curve discrete logarithm problems. Quantum computers, if they become sufficiently powerful, could solve these problems efficiently and break SNARK-based systems. zkSTARKs, which use hash-based cryptography, are considered post-quantum secure, but they have their own trade-offs in proof size and cost.

Complexity of development:

Writing ZKP circuits is much harder than writing regular code. Developers must think in terms of arithmetic circuits, constraint systems, and field elements — concepts that are far removed from everyday programming. This limits the number of developers who can build ZKP-based applications and increases the risk of bugs.

Future directions

Research into ZKPs is advancing rapidly. Several promising directions are being actively explored.

- **Recursive proofs:** Systems like Halo 2 allow a ZKP to verify another ZKP, enabling highly efficient chains of computation that could dramatically reduce the cost of complex privacy-preserving applications.
- **Post-quantum ZKPs:** Researchers are working on proof systems that are secure against quantum computers without sacrificing too much performance.
- **zkEVM (Zero-Knowledge Ethereum Virtual Machine):** Projects are building systems that can prove the correct execution of any Ethereum smart contract using ZKPs, which would enable full privacy and scalability on Ethereum without changing the developer experience.

- **ZKbased machine learning:** Researchers are exploring how ZKPs can be used to prove that a machine learning model was trained correctly on private data, without revealing the training data or the model weights.
- **Standardised ZKP auditing:** As ZKP systems become more widely deployed, the community is developing standard frameworks for auditing ZKP circuits and contracts to catch vulnerabilities before deployment.

By allowing proofs of truth without disclosure of data, ZKPs enable private transactions, private smart contract execution, private identity verification, and scalable computation on public blockchains. Technologies

like zkSNARKs, zkSTARKs, and Bulletproofs each offer different trade-offs, and the right choice depends on the specific requirements of the application.

Significant challenges remain, including computational costs, the risk of trusted setup compromise, implementation vulnerabilities, and the long-term threat of quantum computing. However, research is advancing rapidly, and ZKPs are already being used in production systems like Zcash, zkSync, StarkNet, and Monero.

As open source blockchain ecosystems continue to mature, ZKPs will play an increasingly central role in making these systems both transparent and private — a combination that was once considered impossible but is now becoming a practical reality. 

References

- Goldwasser, S., Micali, S., & Rackoff, C. (1989), The Knowledge Complexity of Interactive Proof Systems. *SIAM Journal on Computing*, 18(1), 186–208, <https://doi.org/10.1137/0218012>
- BenSasson, E., Chiesa, A., Garman, C., Green, M., Miers, I., Tromer, E., & Virza, M. (2014), Zerocash: Decentralized Anonymous Payments from Bitcoin. In 2014 IEEE Symposium on Security and Privacy (SP), pp. 459–474, IEEE, <https://eprint.iacr.org/2014/349>
- Mohanty, H., et al. (2024), A Survey on the Applications of ZeroKnowledge Proofs, arXiv preprint arXiv:2408.00243, <https://arxiv.org/abs/2408.00243>
- Zhou, L., Diro, A., Saini, A., Kaisar, S., & Hiep, P. C. (2024), Leveraging Zero Knowledge Proofs for Blockchain-Based Identity Sharing: A Survey of Advancements, Challenges and Opportunities. *Journal of Information Security and Applications*, 80, 103678, <https://doi.org/10.1016/j.jisa.2023.103678>
- ElHajj, M., & Roelink, B. O. (2024), Evaluating Efficiency of ZKSNARK, ZKSTARK, and Bulletproof in Realworld Scenarios: A Benchmark Study. *Information*, 15(8), 463, <https://doi.org/10.3390/info15080463>
- Roelink, B. O., et al. (2024), Systematic review: Comparing zkSNARK, zkSTARK, and Bulletproof protocols for privacy preserving authentication, *Security and Privacy*, Wiley Online Library, <https://doi.org/10.1002/spy2.401>
- Zhang, B., Pan, H., Li, K., Xing, Y., Wang, J., Fan, D., & Zhang, W. (2024), A Blockchain and Zero Knowledge Proof Based Data Security Transaction Method in Distributed Computing. *Electronics*, 13(21), 4260, <https://doi.org/10.3390/electronics13214260>
- BenSasson, E., Bentov, I., Horesh, Y., & Riabzev, M. (2018), Scalable, Transparent, and PostQuantum Secure Computational Integrity, *Cryptology ePrint Archive*, Report 2018/046, <https://eprint.iacr.org/2018/046>
- Bünz, B., Bootle, J., Boneh, D., Poelstra, A., Wuille, P., & Maxwell, G. (2018), Bulletproofs: Short Proofs for Confidential Transactions and More. In 2018 IEEE Symposium on Security and Privacy (SP), pp. 315–334. IEEE, <https://doi.org/10.1109/SP.2018.00020>
- Tang, X., Shi, L., Wang, X., Charbonnet, K., Tang, S., & Sun, S. (2024), ZeroKnowledge Proof Vulnerability Analysis and Security Auditing, *Cryptology ePrint Archive*, <https://eprint.iacr.org/2024/514>

By: Dr Mohit

The author is a CEH and ECSA, and has worked at IBM, Teramatrix and Sapient. He has written the books titled 'Python Penetration Testing Essential', 'Python: Penetration Testing For Developers', and 'Learn Python in 7 Days'.